ClipSAT : Math for AI & Data Science

Unit 2.01: Neural Layer Matrices & Activations

Name: _____

Q1. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{10} to mathbf{y} in mathbb{R}^{5} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 55
[ ] B: 50
[ ] C: 15
[ ] D: 65

Q2. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{20} to mathbf{y} in mathbb{R}^{10} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 30
[ ] B: 200
[ ] C: 210
[ ] D: 230

Q3. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{30} to mathbf{y} in mathbb{R}^{15} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 45
[ ] B: 450
[ ] C: 465
[ ] D: 495

Q4. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{40} to mathbf{y} in mathbb{R}^{20} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 60
[ ] B: 800
[ ] C: 820
[ ] D: 860

Q5. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{50} to mathbf{y} in mathbb{R}^{25} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 1275
[ ] B: 1250
[ ] C: 75
[ ] D: 1325

Q6. A fully-connected dense layer maps an input mathbf{x} in mathbb{R}^{60} to mathbf{y} in mathbb{R}^{30} using mathbf{y} = mathbf{W}mathbf{x} + mathbf{b} . How many total learnable parameters are in this layer?

[ ] A: 1830
[ ] B: 1800
[ ] C: 90
[ ] D: 1890

Q7. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -2 and z_2 = 3 .

[ ] A: fz_1 = 2, ; fz_2 = 3

[ ] B: fz_1 = -2, ; fz_2 = 3

[ ] C: fz_1 = 0, ; fz_2 = 0

[ ] D: fz_1 = 0, ; fz_2 = 3

Q8. Evaluate the output of the standard ReLU activation function fz = max0, z for inputs z_1 = -4 and z_2 = 6 .

[ ] A: fz_1 = 4, ; fz_2 = 6

[ ] B: fz_1 = -4, ; fz_2 = 6

[ ] C: fz_1 = 0, ; fz_2 = 0

[ ] D: fz_1 = 0, ; fz_2 = 6